

**INSTITUTE FOR ADVANCED AI RESEARCH**

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**NEURAL ARCHITECTURE SEARCH**

*A Comprehensive Survey of Modern Approaches*

**RES-2026-001**

|                          |                                  |
|--------------------------|----------------------------------|
| <b>RESEARCH AREA:</b>    | Machine Learning                 |
| <b>PAPER TYPE:</b>       | Survey                           |
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| <b>INSTITUTION:</b>      | IAIR                             |
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| <b>CITATION COUNT:</b>   | Pending                          |

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**PRINCIPAL INVESTIGATOR**

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# ABSTRACT

This survey provides a comprehensive overview of Neural Architecture Search (NAS) methods, covering evolutionary algorithms, reinforcement learning approaches, and gradient-based methods. We analyze 50+ recent papers and provide insights into future research directions.

# KEY FINDINGS

- 1. Gradient-based methods show 10x speedup over evolutionary approaches
- 2. Transfer learning significantly improves NAS efficiency
- 3. Hardware-aware NAS is becoming the standard for deployment

# SYSTEM STATUS

# SYSTEM STATUS REPORT

Report Date: 2026-01-12T23:54:18.055620

# GIT REPOSITORY STATUS

## Branch Information

Current Branch: feature/campaign-session-binder-system

Commits Ahead: 0

Commits Behind: 0

## Uncommitted Changes

Total Uncommitted Files: 312

Staged Files: 278

Unstaged Files: 311

 **WARNING**

Large number of uncommitted files detected. Consider committing changes.

Recent Commits

- 68541db chore: bump version to 0.7.1
- 8480737 chore: bump version to 0.7.0
- c27fd71 feat: Create FlexiblePDFGenerator for evolving formatting ideas
- 48ae62f docs: Add cognitive processing entry to devlog
- abdfb83 docs: Deep cognitive processing and assumption validation

WORK EFFORTS STATUS

Work Efforts Breakdown

| Category           | Count |
|--------------------|-------|
| Total Work Efforts | 53    |
| Active             | 0     |
| Recent             | 53    |

PROJECT HEALTH METRICS

Health Check Results

| Component           | Status    | Details                    |
|---------------------|-----------|----------------------------|
| _pyrite Structure   | ✔ Valid   | Structure intact           |
| Directory Structure | ✔ Valid   | active/ and backlog/ exist |
| Dependency Lock     | ✔ Present | uv.lock file exists        |

RECENT ACTIVITY

Devlog Entries: 1 recent entries

# ANALYSIS

## STATUS SUMMARY

- Git repository is synchronized
- Project structure is healthy
- Work effort activity: 0 active efforts
- Uncommitted changes: 312 files